



A Survey on Small Language Models in the Era of Large Language Models: Architecture, Capabilities, and Trustworthiness

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Abstract

Large language models (LLMs) based on Transformer architecture are powerful but face challenges with deployment, inference latency, and costly fine-tuning. These limitations highlight the emerging potential of small language models (SLMs), which can either replace LLMs through innovative architectures and technologies, or assist them as efficient proxy or reward models. Emerging architectures such as Mamba and xLSTM address the quadratic scaling of inference with window length in Transformers by enabling linear scaling. To maximize SLM performance, test-time compute scaling strategies reduce the performance gap with LLMs by allocating extra compute budget during test time. Beyond standalone usage, SLMs could also assist in LLMs via weak-to-strong learning, proxy tuning, and guarding, fostering secure and efficient LLM deployment. Lastly, the trustworthiness of SLMs remains a critical yet underexplored research area. However, there is a lack of tutorials on cutting-edge SLM technologies, prompting us to conduct one.

CCS Concepts

• Computing methodologies → Natural language generation.

Keywords

Small Language Models, Weak-to-strong, Trustworthiness

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1 Introduction

Large language models (LLMs) have revolutionized various fields such as AI for science [7, 67, 94, 102], programming [84], and human-centered interaction [121]. However, their massive parameter sizes and commonly used Transformer architecture often pose significant challenges: (1) common local devices cannot load the large-scale parameters and hard to handle the cache storage; (2) huge computational amount cause large inference latency unsuitable for real-time applications; and (3) they make domain-specific fine-tuning computationally demanding. Consequently, small language models (SLMs), which excel in efficiency, cost, and flexibility, have emerged with new potential in the era of LLMs, offering support or replacing LLMs to overcome these challenges. We collect a series of newly released SLMs in Table 1.

The Transformer architecture [93], driven by self-attention, remains the preferred framework for LLMs. To enable local deployment and enhance performance, smaller-scale Transformers are being explored [64, 90] by optimizing components including activation functions, attention mechanisms, layer normalization, and parameter reuse. Despite its strengths, the Transformer's reliance on historical key-value pairs causes inefficient inference, facing quadratic computational complexity $O(L^2)$ and high KV cache memory demands for long sequences. To address these issues, ongoing research is developing new architectures like the Mamba series [20, 22, 28, 30, 50, 115] and xLSTM [13] to improve inference efficiency and memory usage.

Deploying SLMs often results in inferior performance on specific tasks compared to LLMs. To bridge performance gaps, test-time compute scaling [61, 85, 104] proves effective by providing additional computation during inference, such as repeated sampling and verifier-based selection, significantly boosting model performance. Beyond simply replacing LLMs, SLMs could assist

in enhancing their usage and security via weak-to-strong learning [16, 55, 70, 86, 113, 129], proxy tuning [60, 71, 122], decoding [56, 84, 128], and guard [45, 48]. During weak-to-strong learning, SLMs generate weakly supervised datasets that activate LLMs' knowledge. They also function as proxies for fine-tuning by integrating log probabilities with behavioral deltas from smaller models and serve as reward models to guide LLMs during decoding without additional training. Additionally, SLMs enhance LLM security by filtering out potential attack samples, thus mitigating safety risks in conversational AI. Importantly, it is essential to address trustworthiness issues like susceptibility to adversarial attacks and privacy breaches [21, 38, 95]. To summarize, our key contributions include:

- In Section 2, we explore various techniques for enhancing Transformers for SLMs and introduce new architectures suitable for small models.
- In Section 3, we examine strategies for elevating SLMs from weak to strong through test-time scaling and discuss how weak SLMs can support strong LLMs in fine-tuning, decoding, and guarding.
- In Section 4, we assess the trustworthiness of SLMs, focusing on issues including robustness, toxicity, misinformation, hallucination, sycophancy, privacy, and fairness, while providing a comprehensive taxonomy of current evaluation methodologies.

2 SLM Architecture

SLMs commonly employ the Transformer architecture [93], which utilizes self-attention mechanisms to manage long-range text dependencies. Due to the quadratic-time inference associated with the attention mechanism, several subquadratic-time architectures, such as Mamba [30] and xLSTM [13], have been proposed. We have collected newly-released SLMs based on these architectures in Table 1. These architectures are detailed below.

2.1 Transformer for SLMs

Transformer architecture [93] remains the dominant framework for LLMs, utilized across platforms from open-source like Llama [91] to proprietary systems like GPT-4 [3]. Given the extensive use of LLMs, considerable efforts are focused on enhancing Transformer-based SLMs to closely match the performance of their larger counterparts. Generally, in the Transformer architecture, input tokens receive token embeddings and are processed through self-attention that dynamically weights input importance; this output feeds into a feed-forward neural network with an activation function for non-linear processing, and each output is normalized by layer normalization to stabilize learning and facilitate smooth gradient flow. This section examines how Transformers advance SLMs, focusing on core components, including the attention mechanism, activation functions in feed-forward networks, and layer normalization, as well as their strategic parameter sharing.

Self-Attention Mechanism. The self-attention mechanism enables models to assess the significance of all preceding token representations when encoding a current token. It is mathematically expressed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (1)$$

Table 1: Overview of New SLMs (<7B) in 2024 (Excluding Mamba-1, Released in 2023). We correct several erroneous activations in the SLMs collection in [66]. Architecture Symbols: $\mathcal{T}$: Transformer, $\mathcal{M}$: Mamba, $\mathcal{X}$: xLSTM.

Model	Size	Date	Attention	Activation	Layer Norm
$\mathcal{T}$: Phi-4-mini [2]	3.8B	2025.03	GQA	GEGLU	RMSNorm
$\mathcal{T}$: SmolLM-2 [9]	135M, 360M, 1.7B	2025.02	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: DeepSeek-R1-Distill-Qwen-1.5B [32]	1.5B	2025.01	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: PhoneLM [116]	0.5B, 1.5B	2024.11	MHA	ReLU	RMSNorm
$\mathcal{M}$: HyMba [22]	1.5B	2024.11	SSM+TF	GEGLU	LayerNorm
$\mathcal{T}$: MiniCPM-3 [41]	4B	2024.09	MLA	UNK	RMSNorm
$\mathcal{T}$: Llama-3.2 [5]	1B, 3B	2024.09	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: Qwen-2.5 [112]	0.5B, 1.5B, 3B	2024.09	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: Phi-3.5-mini [1]	2.7B	2024.09	GQA	GEGLU	RMSNorm
$\mathcal{T}$: DCLM [54]	1.4B	2024.08	MHA	SwiGLU	LayerNorm
$\mathcal{T}$: H2O-Danube3 [77]	0.5B, 4B	2024.07	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: Fox-1 [89]	1.6B	2024.07	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: SmolLM [10]	135M	2024.07	GQA	SwiGLU	RMSNorm
	360M		GQA	SwiGLU	RMSNorm
	1.7B		MHA	SwiGLU	RMSNorm
$\mathcal{T}$: Minitron [73]	4B	2024.07	MHA	GELU	LayerNorm
$\mathcal{T}$: Gemma-2 [88]	2B	2024.07	GQA	GELU	RMSNorm
$\mathcal{T}$: Qwen-2 [111]	1.8B, 4B	2024.06	GQA	SwiGLU	RMSNorm
$\mathcal{M}$: Zamba [28]	1.2B	2024.05	SSM+TF	UNK	LayerNorm
$\mathcal{X}$: xLSTM [13]	125M, 350M, 760M, 1.3B	2024.05	LSTM	GELU	LayerNorm
$\mathcal{M}$: Mamba-2 [20]	2.7B	2024.05	SSM	SwiGLU	LayerNorm
$\mathcal{T}$: OpenELM [69]	270M, 450M, 1.1B, 3B	2024.04	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: OLMo [29]	1.2B	2024.04	MHA	SwiGLU	RMSNorm
$\mathcal{T}$: Phi-3-mini [1]	3.8B	2024.04	GQA	GEGLU	RMSNorm
$\mathcal{T}$: MiniCPM [41]	1B, 2B	2024.04	GQA	UNK	RMSNorm
$\mathcal{T}$: MobiLlama [90]	0.5B, 1B	2024.02	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: Gemma-1 [87]	2B	2024.02	MQA	GELU	RMSNorm
$\mathcal{T}$: Qwen-1.5 [12]	0.5B	2024.02	GQA	SwiGLU	RMSNorm
$\mathcal{T}$: StableLM-2-zephyr [14]	1.6B	2024.01	MHA	SwiGLU	LayerNorm
$\mathcal{M}$: Mamba 1 [30]	130M, 370M, 790M, 1.4B, 2.8B	2023.12	SSM	SwiGLU	LayerNorm

where $\mathbf{Q} = \mathbf{XW}^Q$, $\mathbf{K} = \mathbf{XW}^K$, and $\mathbf{V} = \mathbf{XW}^V$ denote the query, key, and value matrices, respectively. These matrices are scaled by $\sqrt{d_k}$ to ensure stability, with d_k denoting the dimension of the key matrices. The dot product $\mathbf{QK}^T$ measures the similarity between the query and key vectors. For encoding the current position, the query might be represented as a vector $\mathbf{q}$, resulting in an output that is a weighted vector derived from prior values corresponding to previous token representations. **Multi-Head Attention (MHA)** leverages multiple “heads” to capture diverse information from various representation subspaces. Each head in the Multi-Head Attention mechanism operates independently, allowing the model to process information across different subspaces. This captures a broader range of data features.

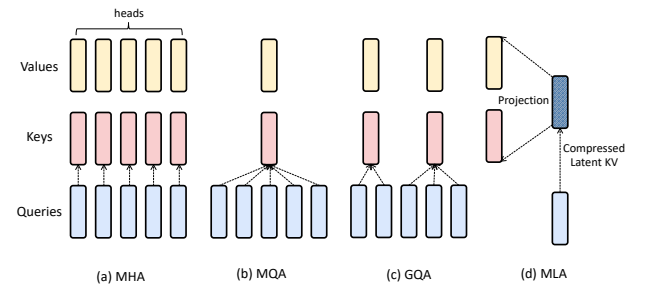


Figure 1: Multi-head Attention Mechanism and its variants.

Table 2: Comparison of Activation Functions.

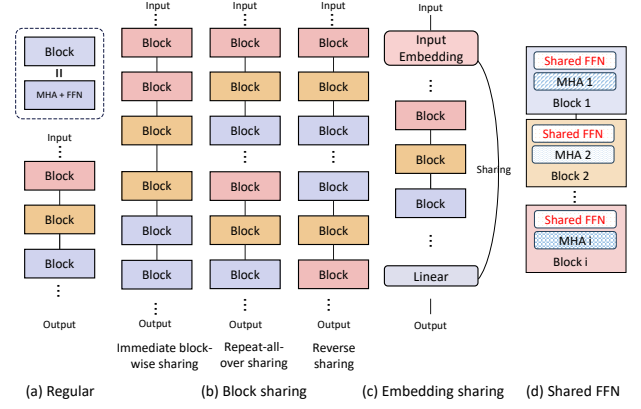
Name	Activation Function
ReLU [4]	$f(x) = \max(0, x)$
GELU [36]	$x \cdot \frac{1}{2} \left[1 + \operatorname{erf} \left(\frac{x}{\sqrt{2}} \right) \right]$
GEGLU [83]	$x \cdot \operatorname{GELU}(Wx + b)$
SwiGLU [83]	$\operatorname{Swish}(x \cdot W + b) \odot (x \cdot V + c), \operatorname{Swish}(x) = x \cdot \operatorname{sigmoid}(x)$

Building on this, many variants have improved the self-attention mechanism for memory efficiency and computational speed including MQA [82], GQA [6], and MLA [59], as shown in Figure 1. Multi-Query Attention (MQA) [82] addresses the KV cache bottleneck in MHA by proposing that all heads share the same keys and values, thus reducing memory and computational overhead. Grouped Query Attention (GQA) [6] strikes a balance by assigning subgroups of query heads to share a single key and value head, minimizing the number of key-value pairs. In contrast, Multi-Head Latent Attention (MLA) [59] compresses keys and values into a latent vector, enhancing management and boosting inference efficiency. *Recent SLMs often utilize GQA in their self-attention mechanisms, offering a flexible balance between reducing cache space and maintaining functionality.*

Feedforward Network (FFN). FFN comprises two linear transformations separated by a non-linear activation function, typically represented as: $\operatorname{FFN}(\mathbf{x}) = \sigma(\mathbf{x}\mathbf{W}_1 + b_1)\mathbf{W}_2 + b_2$, where $\mathbf{W}_1$ and $\mathbf{W}_2$ are the weight matrices, and b_1 and b_2 are bias terms. σ introduces non-linearity, allowing the model to learn complex patterns. Popular activation functions include ReLU [4], GELU [36], GEGLU [83], and SwiGLU [83], shown in Table 2. *ReLU promotes sparsity, which speeds up computation, while SwiGLU offers parameterized flexibility for various tasks, making it favored in SLMs for its effectiveness.*

Layer Normalization. Layer normalization [49] enhances training stability by normalizing the outputs of layers, thereby accelerating convergence. There are primarily two types of layer normalization techniques utilized: (i) **Non-Parametric Layer Norm**, which normalizes inputs based on the mean and variance calculated across the dimensions of a layer without any learnable parameters. (ii) **Parametric Layer Norm**, which incorporates learnable parameters, to allow for adaptive scaling and bias, thus enhancing the model’s flexibility. In addition, **RMS Norm (Root Mean Square Layer Normalization)** [120] simplifies computations by utilizing the root mean square of the inputs. *Owing to its robustness in expressiveness, RMS Norm has gained popularity over traditional Layer Norm in small language models.*

Parameter Sharing. In SLM architectures, parameter-sharing techniques are crucial for reducing model size and space usage. As shown in Figure 2, parameter sharing techniques can be classified into three categories: (i) block sharing, which includes immediate block sharing, repeat-all-over sharing, and reverse sharing, (ii) embedding sharing where the input embedding and the final output weight are identical, and (iii) FFN sharing that involves using a common FFN module across all transformer layers. These approaches can be combined for enhanced efficiency. Repeat-all-over sharing generally performs the best in block sharing, whereas immediate block sharing offers memory savings due to the efficient use of shared cache among closely positioned blocks. Embedding sharing

**Figure 2: Parameter sharing technologies.**

can reduce parameters by about 20% in SLMs, and FFN sharing can achieve a 60% reduction, significantly enhancing parameter efficiency while presenting a viable performance-parameter trade-off.

2.2 Mamba

Transformer architecture suffers from quadratic computational complexity. Mamba series [20, 22, 28, 30, 50, 115] aims to improve inference efficiency and memory usage.

2.2.1 From SSM to Mamba. State space models [31] utilize the minimal number of variables to describe a dynamic system and can process sequential data as follows:

$$\mathbf{h}_k = \bar{\mathbf{A}}\mathbf{h}_{k-1} + \bar{\mathbf{B}}\mathbf{x}_k, \quad \mathbf{y}_k = \mathbf{C}\mathbf{h}_k \quad (2)$$

where $\mathbf{x}_k$ and $\mathbf{y}_k$ are the input-output sequence pairs, $\bar{\mathbf{A}}$, $\bar{\mathbf{B}}$, and $\mathbf{C}$ are learnable weight matrices, and $\mathbf{h}_k$ is the hidden state at time step k . However, traditional SSMs have fixed parameters and treat all input tokens uniformly, which constrains their utility in language modeling. To overcome this limitation, two key innovations transform SSMs into the Mamba architecture:

(1) **Selective Scanning Mechanism.** Traditional SSMs cannot perform selective copying or induction head tasks due to their Linear Time Invariant (LTI) nature, with fixed parameters $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, Δ (where Δ denotes the discrete step size in SSMs) for all input tokens. This design inhibits content-aware inference, essential for language models, as it treats each token uniformly. To overcome this, Mamba introduces input-dependent parameters $\mathbf{B}(x)$, $\mathbf{C}(x)$, $\Delta(x)$, allowing a unique set of $\mathbf{B}$, $\mathbf{C}$, Δ for each input token ($\mathbf{A}$ remains unchanged).

(2) **Hardware-Aware Algorithm.** Frequent data transfers between GPUs’ DRAM and SRAM reduce computational efficiency. To address this issue, a hardware-aware algorithm is designed to minimize these transfers by integrating discrete steps, selective scanning, and multiplication with $\mathbf{C}$ into a single kernel fusion operation.

2.2.2 Mamba Blocks. Multiple Mamba blocks can be stacked and used sequentially, similar to transformer layers, as depicted in Figure 3. Initially, input tokens are linearly embedded into a hidden space, then processed by convolutional layers coupled with the SiLU activation function. Subsequently, a selective SSM module efficiently processes these embeddings to capture context-specific

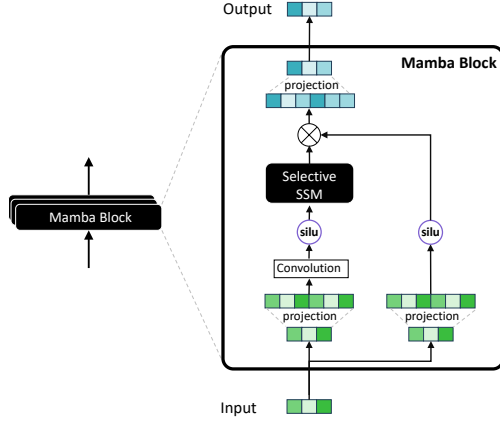


Figure 3: Mamba 1 architecture [30].

information. The resulting representations undergo additional non-linear activation for enhanced expressivity, and a final linear projection maps the features into output space.

2.2.3 Mamba-Derivative Work. Mamba architecture has inspired various innovations focused on fusing Transformer and SSM. Mamba-2 [20] utilizes the structured state space duality to increase computational efficiency, while Jamba [50] alternates attention and SSM layers. In lightweight design, Hymba [22] combines attention and SSM heads in small models (1.5B) for enhanced memory efficiency, and Zamba [28] utilizes a Mamba backbone and a shared attention module to reduce memory use. LongMamba [115] improves long-context capabilities with a token filtering mechanism, enhancing document comprehension. Overall, these initiatives utilize Mamba’s linear complexity to enhance efficient sequential modeling.

2.3 xLSTM

Long Short-Term Memory (LSTM) [37], similar in concept to Mamba [30] for its introduction of time-dependent weights, was pivotal in establishing foundational techniques for language modeling. However, the advent of Transformers [93] has marked a substantial paradigm shift in this field. This raises an intriguing question: Can LSTMs be effectively scaled to billion-parameter models with contemporary technologies? In response, the Extended Long Short-Term Memory (xLSTM) framework [13] has been developed to augment traditional LSTMs by incorporating exponential gating and innovative memory structures. This framework is bifurcated into two distinct components: sLSTM, which utilizes memory mixing for enhanced data integration, and mLSTM, which eschews memory mixing to facilitate parallel processing. Together, these elements comprise the comprehensive xLSTM architecture, as depicted in Figure 4 and further detailed below.

2.3.1 LSTM. LSTMs mitigate the vanishing gradient problem of RNNs through the constant error carousel and gating mechanisms

by the following update rules at time step t :

$$\begin{aligned}
 c_t &= f_t \odot c_{t-1} + i_t \odot z_t & (\text{cell state}) \\
 h_t &= o_t \odot \tilde{h}_t, & \tilde{h} = \psi(c_t) \quad (\text{hidden state}) \\
 z_t &= \varphi(\tilde{z}_t), & \tilde{z}_t = w_z^T x_t + r_z h_{t-1} + b_z \quad (\text{cell input}) \\
 i_t &= \sigma(\tilde{i}_t), & \tilde{i}_t = w_i^T x_t + r_i h_{t-1} + b_i \quad (\text{input gate}) \\
 f_t &= \sigma(\tilde{f}_t), & \tilde{f}_t = w_f^T x_t + r_f h_{t-1} + b_f \quad (\text{forget gate}) \\
 o_t &= \sigma(\tilde{o}_t), & \tilde{o}_t = w_o^T x_t + r_o h_{t-1} + b_o \quad (\text{output gate})
 \end{aligned}$$

where weight vectors w_z, w_i, w_f, w_o correspond to the inputs x_t , and cell, input, forget, and output gates. Recurrent weights r_z, r_i, r_f, r_o link the hidden state h_{t-1} to these gates. Bias terms are b_z, b_i, b_f , and b_o . Activation functions φ (usually $\tanh$) and ψ normalize the otherwise unbounded cell state. All gates use sigmoid activation.

LSTMs have three key limitations: First, they struggle with revising dynamic storage decisions. Second, they have limited storage capacity, as all data is compressed into a single cell state c . Third, their parallel processing capabilities are hindered by memory mixing, necessitating sequential hidden-to-hidden connections.

2.3.2 sLSTM. To improve the ability of LSTMs to revise storage decisions, exponential gates have been introduced. Particularly, input and forget gates can utilize exponential activation functions. For normalization purposes, a normalizer state is introduced that aggregates the product of the input gate with all subsequent forget gates. Based on the traditional LSTM, the sLSTM changes the input gate to $i_t = \exp(\tilde{i}_t)$, the forget gate to $f_t = \sigma(\tilde{f}_t)$ or $\exp(\tilde{f}_t)$, adds a normalizer state $n_t = f_t \odot n_{t-1} + i_t$, and changes the hidden state to $h_t = o_t \odot \tilde{h}_t$, where $\tilde{h}_t = \frac{c_t}{n_t}$.

The sLSTM enables multiple memory cells similar to the original LSTM via recurrent connections from the hidden state h to memory cell inputs z . This setup supports multiple heads without memory mixing across the heads, but only memory mixing across cells within each head.

2.3.3 mLSTM. To improve LSTM storage capacity, the memory cell dimensionality is increased from a vector c to a matrix $C \in \mathbb{R}^{d \times d}$. Based on the traditional LSTM, the mLSTM modifies the cell state equation to $C_t = f_t \odot C_{t-1} + i_t \odot (v_t k_t^T)$, introduces $n_t = f_t \odot n_{t-1} + i_t \odot k_t$, and defines $q_t = W_q x_t + b_q$, $k_t = \frac{1}{\sqrt{d}} (W_k x_t + b_k)$, $v_t = W_v x_t + b_v$. It changes the input gate to $i_t = \exp(\tilde{i}_t)$, the forget

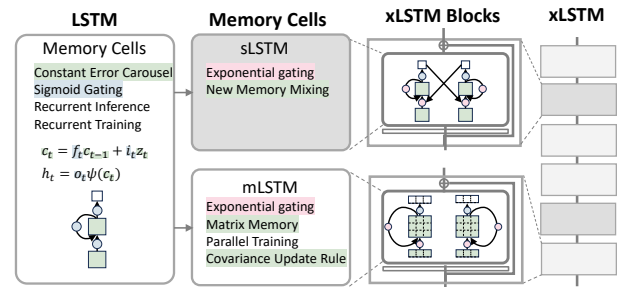


Figure 4: The extended LSTM (xLSTM) family [13].

gate to $\mathbf{f}_t = \sigma(\tilde{\mathbf{f}}_t)$ or $\exp(\tilde{\mathbf{f}}_t)$, and the hidden state to $\mathbf{h}_t = \mathbf{o}_t \odot \tilde{\mathbf{h}}_t$, where $\tilde{\mathbf{h}}_t = \frac{C_t \mathbf{q}_t}{\max\{|\mathbf{n}_t^\top \mathbf{q}_t|, 1\}}$.

2.3.4 xLSTM. xLSTM has two block types: (i) post up-projection, using sLSTM with optional convolutions and gated MLPs, and (ii) pre up-projection, using mLSTM enclosed in two MLPs via convolution, learnable connections, and output gates. xLSTMs are built with these blocks and feature LayerNorm residual backbones. xLSTM is compared with Llama [45], and Mamba [30] across model sizes (125M, 350M, 760M, 1.3B). In the PALOMA benchmark [68], it excels in length extrapolation and outperforms same-size baselines in downstream tasks. xLSTM shows better scalability and higher throughput than Transformers and SSMS due to efficient memory use, allowing larger batches. These findings highlight xLSTM’s potential in small model applications, though scaling to larger models remains unexplored. Due to xLSTM’s unique efficiency in long-term sequence modeling, it has become the preferred architecture for many follow-up studies, including modeling long-range dependencies in biological and chemical sequences [80], time series forecasting [8], audio [110], and stock prediction [25].

3 Weak to Strong Methods

In the era of large language models, small language models maintain advantages due to their easy deployment, despite typically underperforming compared to large language models. To bridge this gap, methods have been developed to enhance SLMs, enabling them to beat LLMs in specific scenarios, noted as "SLMs from weak to strong." Moreover, SLMs’ lightweight nature facilitates their usage in supporting LLMs through fine-tuning, decoding, and safeguarding, illustrating another aspect of the "weak-to-strong" strategy.

3.1 SLMs Beat LLMs in Specific Scenarios

Typically, SLMs underperform LLMs due to smaller parameter sizes. However, developers employ technologies such as supervised fine-tuning, distillation, and quantization to enhance SLMs. Among them, test-time scaling is a cutting-edge technology that can potentially beat LLMs, so we highlight recent work on it. For more details on other technologies enhancing SLMs, see Wang et al. [96].

Test-Time Compute Scaling. Test-time compute scaling (also known as inference scaling) enhances language models by allocating more computational resources during test time, utilizing strategies such as best-of-N [57], majority voting [100], and tree search [108]. A typical best-of-N approach involves sampling multiple outputs from the model, which are then assessed by a reward-based verifier to select the optimal one. Both Snell et al. [85] and Wu et al. [104] explore the trade-off between test-time scaling and training scaling. Snell et al. [85] found that SLMs with test-time compute scaling outperform models that are 14 times larger. Wu et al. [104] indicates that smaller models equipped with advanced test-time scaling algorithms offer superior performance. Further research by Liu et al. [61] shows that under test-time scaling strategies, smaller models excel all while requiring fewer FLOPs, e.g., a 0.5B SLM outperforms GPT-4o, a 3B LLM surpasses a 405B LLM, and a 7B LLM exceeds both o1 and DeepSeek-R1 [32].

Compute-optimal Test-Time Scaling. Compute-optimal test-time scaling aims to allocate the optimal compute for each problem,

being defined as selecting optimal test-time scaling hyperparameters for a given prompt, such as scaling strategies, policy model selection, model size, and prompt designs. Various studies define the compute-optimal test-time scaling strategy, each differing in the selection space of hyperparameters [58, 61, 85]. Snell et al. [85] first formally defines $\text{Target}(\theta, N, q)$ as the distribution over output tokens induced by the language model for a given prompt q , using test-time compute hyper-parameters θ , and a compute budget of N . The goal is to select optimal hyperparameters θ that maximize accuracy for a given problem. Formally:

$$\theta_{q,a^*}^*(q)(N) = \arg \max_{\theta} \left(\mathbb{E}_{y \sim \text{Target}(\theta, N, q)} [\mathbf{1}_{y=y^*(q)}] \right) \quad (3)$$

where $y^*(q)$ is the ground-truth response for q , and $\theta_{q,a^*}^*(q)(N)$ indicates the strategy for problem q within a given budget N . This definition shows that scaling strategies are question-dependent, which slightly diverges from the concurrent study [104]. The study [61] suggests incorporating reward models in scaling decisions.

Strategies for Compute-optimal Scaling. Building on this problem, existing works design specific test-time compute-optimal scaling methods. Snell et al. [85] finds that the efficacy of methods varies with the computational budget and problem difficulty: Beam search excels with complex problems and limited compute budget; best-of-N is more effective for simpler problems and higher budgets. Inspired by this, they adapt search settings to take into account problem difficulty and compute budget, significantly enhancing performance. Additionally, Liu et al. [61] further refines this method to account for variability in reward models, as they perform differently at various inference lengths. For instance, tokens scaled with RLHFlow-PRM-Deepseek-8B are consistently larger than those of RLHFlowPRM-Mistral-8B—nearly double—owing to the longer training data length in DeepSeek PRM compared to Mistral-PRM-Data.

3.2 SLMs Help LLM Fine-tuning

This subsection introduces two paradigms by which SLMs can assist in fine-tuning of LLMs: **weak-to-strong learning** and **proxy fine-tuning**. *First, weak-to-strong learning can guide LLMs with SLM-generated datasets.* As LLMs advance and often outperform humans in complex tasks, the challenge of providing high-quality data emerges due to potentially simplistic or incorrect human annotations [16, 55, 113, 123, 129]. Burns et al. [16] introduced the *weak-to-strong learning* to explore if weak SLMs can effectively guide strong LLMs. This method is effective for two reasons: (1) strong models can replicate weak models, including their errors, while maintaining robust task representations, and (2) even inaccurate supervision from weak models can trigger pre-existing knowledge or capabilities. Strong models trained with weak supervision often underperform those fine-tuned on ground-truth data. To bridge this gap, Yang et al. [113] and Zhou et al. [129] focus on increasing data utilization effectiveness from weak models, employing strategies such as contrastive learning on incorrect samples and intensively learning on samples where LLMs are overconfident. SLMs could also enhance data quality. For example, Superfiltering [55] leverages SLMs to filter low-quality data for LLMs via difficulty-aware selection. *Second, proxy fine-tuning with SLMs can approximate gradients for fine-tuning large-scale LLMs on target*

datasets [60, 71, 122, 124]. Fine-tuning LLMs is costly, while SLMs can effectively approximate the gradients required for fine-tuning LLMs. For instance, both Proxy-Tuning [60] and Emulated Fine-Tuning (EFT) [71] fine-tune a smaller LM and apply the predictive differences from the tuned and untuned small LMs to modify the original predictions of a larger, untuned model, while maintaining the benefits of large-scale pretraining. In a related direction, Lo-RAM [122] employs LoRA on a pruned SLM, obtaining incremental low-rank matrices that are subsequently restored onto the original large model during inference. Zhang et al. [124] propose training LLM agents without modifying weights by forging agent functions, which can potentially be implemented using small language models.

3.3 SLMs Guide LLM Decoding

Learning from human feedback has gained widespread attention due to its ability to utilize human-annotated data to align with human preferences, primarily by maximizing expected reward scores from implicit or explicit reward models. Alternatively, SLMs could serve as reward models for the alignment. Typically, this alignment process involves a greedy search at test time aimed at maximizing the log probability offsets between the tuned and untuned small models while sampling from a fixed large model [130]. Such weak-to-strong search strategy has been applied to various LLM applications, such as safety alignment [84], jailbreak attacks [128], and unlearning [43].

Multi-objective Decoding Considering the different alignment objectives across scenarios and users, there is a need for on-the-fly adaptation of language models to cater to various objective combinations. This raises a question: Given a set of policies corresponding to different reward models, can we find a way that does not require fine-tuning LLMs to correspond to the multi-policies, which would be time-consuming and difficult? Multi-Objective Decoding (MOD) [84] introduces a method for combining prediction distributions based on multiple reward models, each trained for an individual objective. This approach identifies a closed-form solution among f-divergence regularized alignment methods like PPO and DPO through Legendre transform [75]. This method allows for the combination of any reward models at inference time, eliminating the need for retraining reward models.

Weak-to-strong Jailbreak Attack The weak-to-strong search can also conduct jailbreak attacks on LLMs [128], revealing that most jailbreak tokens surface within the first ten tokens. The logits from strong, safe models, plus the unsafe logit offset between weak safe models and unsafe ones, can produce a jailbreak token from the strong model. This suggests that even secure LLMs can be misled by unsafe, weak SLMs into generating undesired outputs with targeted guidance. This method is computationally efficient, avoiding the need for extensive computations to find optimal decoding parameters or optimize prompts, and it can produce more harmful content than smaller attack models alone.

Weak-to-strong Unlearning The weak-to-strong decoding can also be applied to LLM unlearning issues. LLMs may cause privacy leaks and copyright infringement due to their memory of training data. Existing LLM unlearning methods require retraining models from scratch [126], which is impractical for black-box models.

Hence, δ -Unlearning [43] is proposed, an offset unlearning framework for black-box LLMs that learns the required logit offsets by comparing the logits of a small, white-box model with the original logits without fine-tuning the black-box LLM.

3.4 SLMs Guard LLMs

As demonstrated in recent studies [33, 62, 78, 81, 119], LLMs exhibit significant vulnerabilities to adversarial attacks and jailbreaking attempts. For instance, Wang et al. [98] illustrate that ChatGPT underperforms when evaluated on adversarial datasets, highlighting ongoing risks associated with adversarial vulnerabilities. This reinforces the need for robust guardrails in generative AI systems. Beyond developing inherently trustworthy LLMs, the adoption of SLMs for reinforcing LLM safety [45, 48] and hallucination detection [92, 109] has gained considerable attention.

SLMs as Guardian. To address safety concerns of LLMs, several works [45, 48, 78, 99] have been conducted to use SLMs to help improve the safety of LLMs. For example, Llama Guard [45], fine-tuned on Llama2-7B, provides a publicly available tool designed explicitly for identifying safety risks in conversational prompts and responses. However, this tool primarily assesses the harmfulness of questions and answers without enabling the generation of fluent, safeguarded responses. To address this shortcoming, Kwon et al. [48] propose a specialized SLM capable of concurrently detecting harmful queries and generating safeguard-oriented, explanatory responses, thus significantly enhancing conversational AI safety. Additionally, Sawtell et al. [78] illustrates that SLMs can serve as robust feature extractors, effectively training simple classifiers with fewer than 100 high-quality examples. These classifiers support tasks including content safety classification, prompt injection detection, and simultaneous token generation for improved safety. Further advancements include STAND-Guard [99], which fine-tunes SLMs to monitor the safety of generated content, especially addressing out-of-distribution scenarios. Similarly, TorchOpera [34] integrates multiple SLMs serving as safety detectors and repair modules, enhancing prompt safety and response quality through enriched context and defined corrective guidelines.

SLMs as Hallucination Detectors. To address the challenge of hallucination detection in LLMs, recent works [11, 92, 109, 127] have focused on calibrating model confidence through auxiliary SLMs. These auxiliary models analyze both the original questions and the corresponding answers produced by LLMs to generate reliable confidence estimates. The calibration training minimizes discrepancies between predicted confidence scores and actual calibration errors. For example, APRICOT [92] leverages an auxiliary DeBERTaV3 model [35] to assess the confidence of LLM responses to improve the expression of uncertainty and the precision of response adjustments. Similarly, POLAR [127] proposes a self-supervised calibration technique to align LLM outputs with various weak supervision signals, which refines model confidence and reducing potential inaccuracies. Moreover, SLMs have also been employed to evaluate the internal states of LLMs, predicting the probability of hallucinated content. For example, Xu et al. [109] introduce a lightweight detector to analyze token-level contributions to hallucinations. SAPLMA [11] demonstrates that internal states within LLMs carry valuable signals about the truthfulness of their generated statements and

achieves impressive classification accuracies ranging from 71% to 83%. Additionally, recent approaches [33, 40] have integrated SLMs into unified frameworks designed to enhance real-time hallucination detection. For instance, Hu et al. [42] propose using an SLM classifier for initial hallucination detection. Furthermore, Han et al. [33] employs customized SLMs to identify unsafe or hallucinated content for LLMs. Collectively, these approaches demonstrate the critical role of SLMs in fortifying LLM safety, emphasizing proactive risk mitigation and enhancing overall AI safety and user trust.

4 Trustworthiness in SLMs

Despite their widespread use, small language models pose risks in adversarial robustness, toxicity, privacy, and fairness. For a broader overview, we refer readers to see Wang et al. [96]. Below, we highlight recent advances in these areas for small language models.

Adversarial Robustness. Adversarial robustness in LMs refers to a model’s ability to resist malicious inputs designed to manipulate its behavior or degrade its performance [63]. Recent studies have demonstrated that small language models are particularly vulnerable to various forms of adversarial attacks. These include malicious in-context demonstrations [72], adversarial word replacement [114], and adversarial suffix tokens [39]. However, there is currently no consensus on the relationship between model size and adversarial robustness. For instance, Yang et al. [114] observed that larger models exhibit improved robustness under adversarial word replacement, whereas other studies [39, 72] report different trends across different attack types. To tackle the adversarial attacks, several adversarial defense techniques have been proposed [47, 63, 95, 106, 118]. Yu et al. [118] introduced an adversarial training framework to enhance model robustness, and Xhonneux et al. [106] extended this work to defend against continuous adversarial perturbations. Furthermore, certifiably robust LMs have been proposed [47, 95], offering formal guarantees against specific classes of adversarial prompts. Howe et al. [39] found adversarial training more effective on larger models, implying greater training effort should be allocated to smaller models. These findings underscore **the need for increased defensive computing and thorough adversarial robustness evaluation in small language models to ensure their reliability in real-world applications.**

Toxicity and Refusal Behaviors. Toxicity in LMs refers to the generation of harmful, offensive, or inappropriate content that may cause harm to individuals or groups [97]. Modern language models are expected not only to avoid generating toxic outputs but also to actively refuse to respond to harmful or unsafe prompts.

Recent studies systematically benchmark toxicity and refusal behaviors in language models ranging from SLMs to LLMs [18, 24, 74, 101, 107]. A consistent finding across these studies is that *model size does not correlate strongly with safety performance in toxicity mitigation*. For instance, LLaMA-2 7B exhibits the highest refusal rate on harmful prompts in the Do-Not-Answer dataset [101], while in OR-Bench [18], LLaMA-3 8B demonstrates safer behavior than both LLaMA-3 70B and GPT-3.5-turbo-0125. Notably, Cui et al. [18] emphasize a trade-off between helpfulness and response safety. Similarly, SORRY-Bench [107] highlights that refusal performance varies significantly across models and fine-grained categories. A concerning trend emerges with on-device SLMs, particularly those

around 3B parameters, such as Gemma-2B, and especially after aggressive quantization (e.g., 4-bit). These models often generate toxic, hateful, or illegal outputs without requiring jailbreak techniques. This vulnerability may result from quantization masking or removing safety layers, or from the lack of explicit refusal mechanisms in lightweight SLMs. These observations further reinforce that model size alone is not a reliable predictor of toxicity or safety in refusal behavior.

Several mitigation strategies address toxicity across different stages of LM development. During pretraining and fine-tuning, filtering toxic data [46, 65] and training on curated, safe datasets using techniques such as reinforcement learning from human feedback (RLHF) [105] show promise. At inference time, methods like contrastive prompting help steer generation away from harmful content [51]. In the post-processing phase, auditing tools and classifiers detect and filter toxic outputs before deployment [103]. Given the increasing deployment of SLMs in resource-constrained settings, future research should prioritize **toxicity mitigation techniques tailored to on-device models, especially under quantization.**

Jailbreak attacks, in adversarial settings, aim to craft sophisticated prompts that bypass a model’s safety mechanisms, thereby coercing the model into responding to malicious queries and generating harmful or toxic content. Small language models often prioritize helpfulness over harmlessness, making them particularly susceptible to such attacks. Zhang et al. [125] conduct a comprehensive evaluation of SLMs under jailbreak attacks, assessing 63 models across 8 distinct attack strategies. Their findings reveal that approximately half of the evaluated SLMs are highly vulnerable to jailbreak prompts. Importantly, the study shows that model size has little correlation with jailbreak vulnerability, while training techniques play a critical role in determining the security posture of SLMs. These observations underscore **the importance of incorporating robust safety mechanisms during the design and training phases of SLM development.**

Privacy. Privacy-preserving capabilities of SLMs are essential, as even basic interactions can severely disseminate personally identifiable information (PII) [52, 53], potentially violating major privacy laws like the EU’s General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA). The existing research on SLM privacy primarily focuses on (1) benchmarking SLM privacy issues [53] and (2) privacy-preserving strategies for SLM generation [23, 44, 53, 117]. Regarding evaluating SLMs’ privacy, Li et al. [53] proposes a privacy evaluation benchmark, PrivLM-Bench, to assess the privacy issues in LMs. Three levels of attacks are implemented, including data extraction attacks [17], membership inference attacks [27], and embedding-level privacy attacks [76]. Their experimental results suggest that LMs have limited privacy-preserving capabilities. Regarding privacy-preserving strategies, research on privacy-preserving LMs (PPLMs) mainly focuses on differential privacy (DP) mechanisms [44, 53, 117]. Duan et al. explored DP prompt tuning by adding noise to soft prompts for enhanced privacy. This remains an active area of research, particularly as more organizations fine-tune small models on proprietary data, raising open questions about how to prevent unintended data exposure. **A promising direction is to integrate differential privacy into**

parameter-efficient tuning (e.g., LoRA), develop quantization-aware privacy controls, and build auditing tools to detect memorization or leakage in on-device settings.

Fairness, Bias, and Stereotype. Fairness in language models refers to the absence of unjust or prejudiced behavior toward specific groups (e.g., based on gender, race, etc.), and more broadly to the avoidance of harmful stereotypes or biased outputs [15]. Small language models (SLMs), like their larger counterparts, inherit biases from their training data. Cui et al. [19] introduce a benchmark focusing on identity, credit, criminality, and health-related questions to assess fairness, and find that 7B SLMs perform significantly worse in these aspects. Nakka et al. [74] study both on-server and on-device SLMs and show that quantized on-device models exhibit higher risks of stereotypical bias and unfair behaviors. AdvCoU [72] reveals that adversarial in-context demonstrations achieve 100% attack success rates on stereotype and sycophancy dimensions across both small and large models. Similar to other trustworthiness dimensions, there is no consistent relationship between model size and fairness vulnerabilities [24]. Some initial efforts aim to improve fairness in SLMs [26]. For example, Fayyazi et al. [26] propose a fairness-aware framework, FACTER, designed for LM-based recommendation systems. FACTER employs an adaptive semantic variance threshold and a violation-triggered tightening mechanism to automatically enhance fairness constraints when biased patterns are detected. It achieves up to a 95.5% reduction in fairness violations on SLMs such as Mistral-7B and LLaMA-2-7B. ROBBIE [24] demonstrates that the effectiveness of debiasing methods varies by model size: self-debiasing [79] is more effective on smaller models, while prompting methods are more effective on larger models. **A key open challenge is ensuring fairness in multilingual SLMs, where biases specific to both cultural contexts and low-resource languages are prevalent yet remain underexplored.**

5 OPEN CHALLENGES AND FUTURE DIRECTIONS

In this section, we discuss the open challenges and several corresponding promising directions to further advance SLM studies with technologies that are already applied in LLMs.

- **Retrieval Augmented Generation for SLMs** Recent advances in Retrieval-Augmented Generation enhance large language model integration of external knowledge, but do not transfer well to small language models, which struggle with complex queries and multi-step reasoning. A specialized RAG paradigm using graph structures offers a promising solution for SLMs by leveraging graph connections and hierarchical representations to reduce cognitive load and enhance reasoning. This approach necessitates developing lightweight graph-based retrieval algorithms and hybrid data systems that integrate text with graph structures.
- **Multi-agent with SLM Collaboration** The SLM collaboration-based multi-agent system is redefining development with its suitability for distributed deployment and improved performance in adaptive collaboration networks. Unlike a single large LLM that demands significant resources, a network of smaller SLMs offers dynamic calling for computational efficiency. This decentralized approach leverages expert models to enhance task execution,

with systems like eight SLMs outperforming a large LLM and providing efficient advanced AI capabilities for edge devices.

- **Advancing Trustworthy SLMs:** Ensuring the trustworthiness of SLMs remains a critical challenge requiring deeper investigation. Key areas of focus include mitigating the generation of toxic content and misinformation, both of which are currently underexplored in SLMs. Additionally, no effective methods have been developed to address sycophancy. Another pressing concern is the design of fairness-aware SLMs that function effectively across diverse domains while minimizing biases, thereby promoting ethical and responsible AI deployment.

6 Conclusion

In this survey, we have reviewed recent advancements in small language models in the era of large language models. We begin by examining architectures tailored for SLMs, including Transformer design, Mamba, and xLSTM. We then explore weak-to-strong methods such as test-time scaling, which enhance SLMs by enabling them to surpass LLMs and assist in the fine-tuning, decoding, and safeguarding of LLMs. Additionally, we address the critical issue of trustworthiness in SLMs, a significant concern in contemporary language models. This survey concludes by offering key insights that will inform and guide future research on small language models.

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